

- 1. Preliminary Analysis
- 2. Frequency Analysis
- 3. Optimization Modeling

Water Resource System Engineering Term Project

Research Group for Climate Change Adaptation in Water Resources (CCAWR)
Department of Civil and Environmental Engineering
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Preliminary Analysis

보령댐 및 보령댐 유역에 대한 예비분석

- 충청남도 보령시에 위치한 보령댐 유역으로 1998년 한국수자원공사가 충청남도 서북부 지역에 생활용수와 공업용수 등을 공급할 목적으로 건설한 다목적댐
- 보령댐의 유역면적은 163.7km², 유로연장 22.3km이며, 유역 평균경사는 40.19%
- 유역 면적의 약 80%는 산림으로 이루어져 있고 약 13%는 논, 나머지 7%는 도심지
- 2015년 우리나라 중부지방을 중심으로 발생한 가뭄으로 2015년 11월 보령댐의 저수율이 18.9%까지 낮아져 준공 이후 최저 저수율을 기록, 이에 충남지역은 제한급수를 시행하고 금강에서 보령댐을 잇는 11.49km 길이의 도수로 시설을 긴급 설치

1. Preliminary Analysis

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구분	내용
시설명	보령
종류	다목적
형식	E.C.R.D
길이	291
높이	50
정상표고	79
여수로 율류포고	64
관할기관	K-water
준공일자	2000.06.30
좌안	충청남도 보령시 미산면 용수리
우안	충청남도 보령시 주산면 동오리
유역면적	163.6
하천	웅천천
계획홍수위	75.5
상시만수위	74
저수위	50
홍수기제한수위	74
총저수용량	116.9
유효저수용량	108.7
홍수조절용량	10

Preliminary Analysis

현재 직면하고 있는 문제

강수량 부족에 보령댐 저수율 31.5%...가뭄 주의 단계 진입

송고시간 | 2021-07-28 15:06



이은파 기자
기자페이지

하천 유지용수 하루 1만3천t 감량...생활·공업·농업용수는 실사용량만큼 공급



충남도 가뭄 위기 '비상' ... 8개 시군 '경계' 발령

최신용 | © 연합뉴스 2022.05.16 16:29 | 댓글 0

이필영 권한대행, 용수공급 사업 적극 추진 주문



충남 보령댐 가뭄 '비상'...충남도, 관계기관과 대책마련 분주

금강도수로 운영기준 개선·용담댐 활용 등 개선안 추진

등록 2021-08-19 오전 9:37:33
수정 2021-08-19 오전 9:37:33

가 가



가뭄 상황 전파 및 용수공급상황실 운영 회의가 열리고 있다. 사진=충남도 제공

보령댐 저수율 평년의 절반... '가뭄 비상'에 금강 물 댈다

중앙일보 | 입력 2019.08.28 00:23

자면보기 ①

신진호 기자

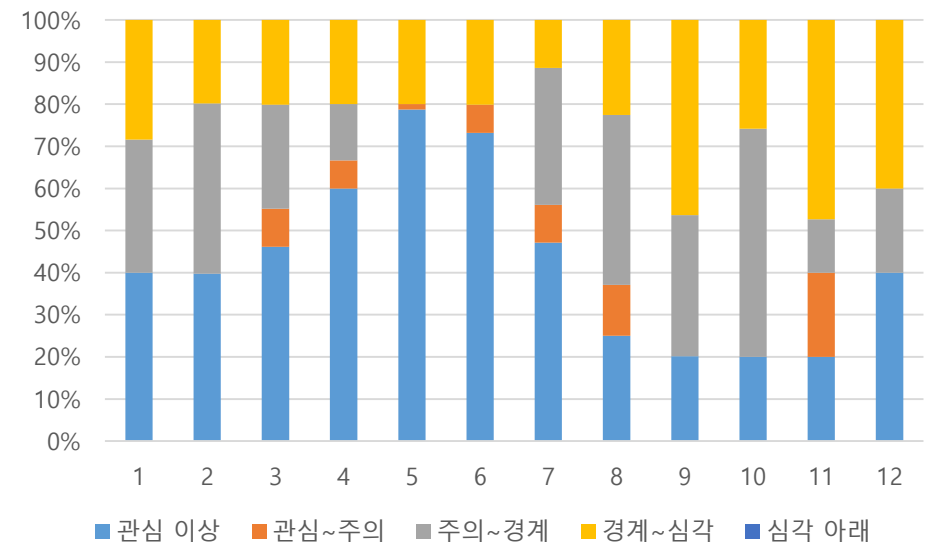
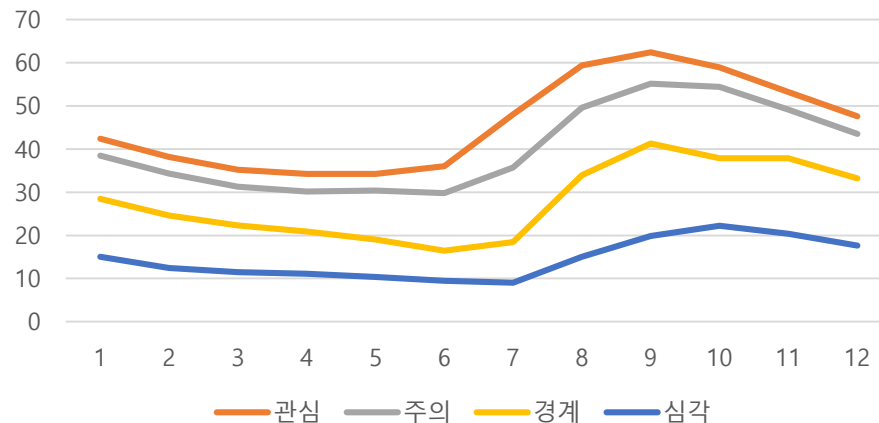
구독



26일 충남 보령댐 저수율이 평년의 절반수준인 28%대로 낮아지면서 충남도와 수자원공사는 원활한 생활·공업용수 공급을 위해 금강-보령댐 구간에 설치된 도수로 가동을 시작했다. [연합뉴스]

보령과 서산·태안·홍성 등 충남 서북부지역에 생활·공업용수를 공급하는 보령댐 수위가 낮아지면서 금강 물을 보령댐으로 공급하는 도수로(물을 댈 도랑)가 가동됐다. 가뭄이 지속하면서 물 부족을 더는 지켜볼 수 없다는 판단에서다.

보령댐 용수공급조정기준



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Preliminary Analysis

개선가능한 부분

- The 23 unsolved problems in hydrology

Table 1. The 23 unsolved problems in hydrology identified by the community process in 2018.

Time variability and change

1. Is the hydrological cycle regionally accelerating/decelerating under climate and environmental change, and are there tipping points (irreversible changes)?
2. How will cold region runoff and groundwater change in a warmer climate (e.g. with glacier melt and permafrost thaw)?
3. What are the mechanisms by which climate change and water use alter ephemeral rivers and groundwater in (semi-) arid regions?
4. What are the impacts of land cover change and soil disturbances on water and energy fluxes at the land surface, and on the resulting groundwater recharge?

Space variability and scaling

5. What causes spatial heterogeneity and homogeneity in runoff, evaporation, subsurface water and material fluxes (carbon and other nutrients, sediments), and in their sensitivity to their controls (e.g. snow fall regime, aridity, reaction coefficients)?
6. What are the hydrologic laws at the catchment scale and how do they change with scale?
7. Why is most flow preferential across multiple scales and how does such behaviour co-evolve with the critical zone?
8. Why do streams respond so quickly to precipitation inputs when storm flow is so old, and what is the transit time distribution of water in the terrestrial water cycle?

Variability of extremes

9. How do flood-rich and drought-rich periods arise, are they changing, and if so why?
10. Why are runoff extremes in some catchments more sensitive to land-use/cover and geomorphic change than in others?
11. Why, how and when do rain-on-snow events produce exceptional runoff?

Interfaces in hydrology

12. What are the processes that control hillslope–riparian–stream–groundwater interactions and when do the compartments connect?
13. What are the processes controlling the fluxes of groundwater across boundaries (e.g. groundwater recharge, inter-catchment fluxes and discharge to oceans)?
14. What factors contribute to the long-term persistence of sources responsible for the degradation of water quality?
15. What are the extent, fate and impact of contaminants of emerging concern and how are microbial pathogens removed or inactivated in the subsurface?

Measurements and data

16. How can we use innovative technologies to measure surface and subsurface properties, states and fluxes at a range of spatial and temporal scales?
17. What is the relative value of traditional hydrological observations vs soft data (qualitative observations from lay persons, data mining etc.), and under what conditions can we substitute space for time?
18. How can we extract information from available data on human and water systems in order to inform the building process of socio-hydrological models and conceptualisations?

Modelling methods

19. How can hydrological models be adapted to be able to extrapolate to changing conditions, including changing vegetation dynamics?
20. How can we disentangle and reduce model structural/parameter/input uncertainty in hydrological prediction?

Interfaces with society

21. How can the (un)certainly in hydrological predictions be communicated to decision makers and the general public?
22. What are the synergies and tradeoffs between societal goals related to water management (e.g. water–environment–energy–food–health)?
23. What is the role of water in migration, urbanisation and the dynamics of human civilisations, and what are the implications for contemporary water management?

- 기후변화로 인한
- 수자원시스템 최적화를 통한 방류량 산정으로 기후변화와 강우에서도 확률의 개념을 사용하여 피해방지를 완화가는 것은 개선 가능

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Frequency Analysis

보령댐이 최근 겪은 가뭄은 얼마나 심각했는가?

- 보령댐의 연최저 저수량을 기준으로 빈도분석을 진행
- 1998~2021 정상성 부적합
- 정상성을 만족하는 1998~2013년으로 빈도분석



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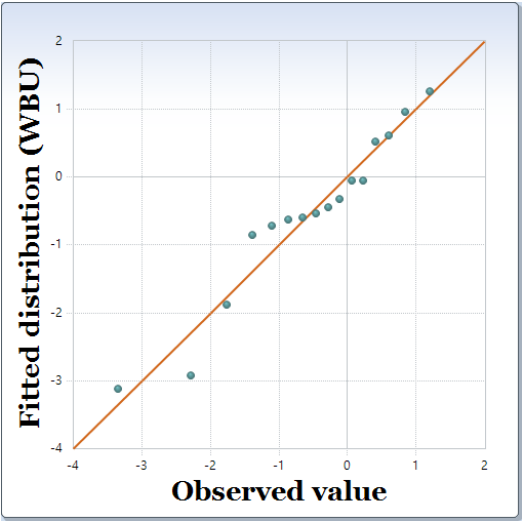
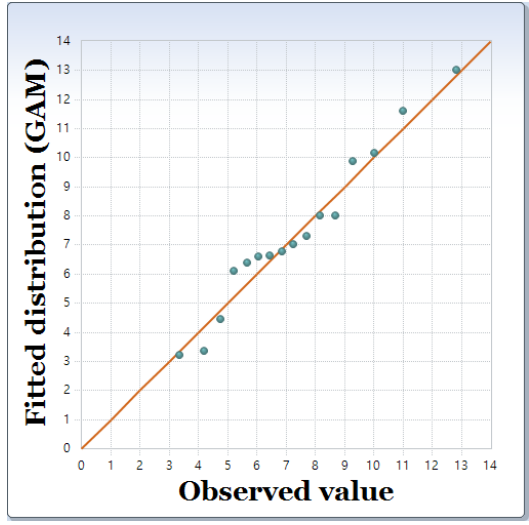
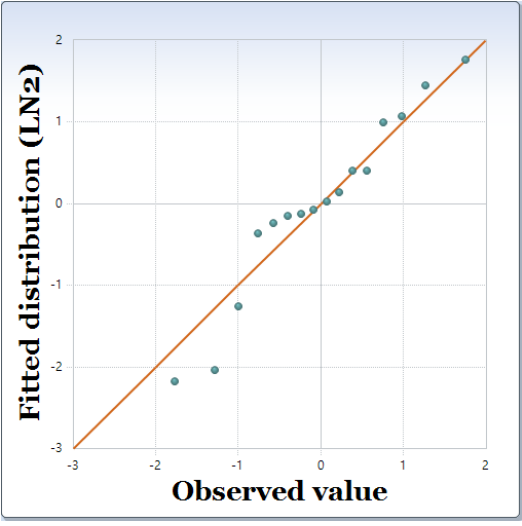
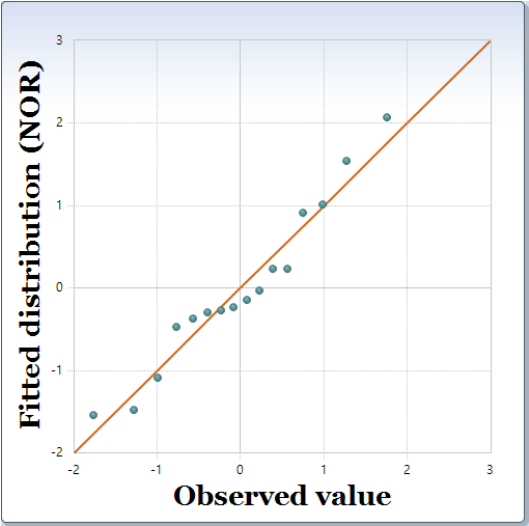
1998~2021	Computed	Table	Accept
Anderson correlogram Test	0.000	0.033	O
Runs Test	1.647	1.960	O
Spearman rank cor. coeff. test	3.611	2.074	X
Turning Point Test	0.671	1.960	O
Wald-Wolfowitz Test	1.213	1.960	O
Mann-Whitney Test	2.309	1.960	X
Mann Kendall Test	2.877	1.960	X
Hotelling-Pabst Test	2.961	1.960	X
t Test	2.969	2.074	X
Sen Test	-0.8	-1.8 ~ -0.4	O

1998~2013	Computed	Table	Accept
Anderson correlogram Test	0.000	0.200	O
Runs Test	0.277	1.960	O
Spearman rank cor. coeff. test	1.378	2.145	O
Turning Point Test	0.420	1.960	O
Wald-Wolfowitz Test	0.118	1.960	O
Mann-Whitney Test	1.785	1.960	O
Mann Kendall Test	1.441	1.960	O
Hotelling-Pabst Test	1.424	1.960	O
t Test	1.372	2.074	O
Sen Test	-1.2	- 3.2 ~ 0.4	O

Frequency Analysis

Parameter Estimation : METHOD OF MOMENTS

Dist.	LOC PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY
NOR	41.01250	15.07377	0.00000	O
LN2	0.00000	3.64052	0.35596	O
LN3	-62.61074	4.63029	0.14471	O
GAM	0.00000	5.54023	7.40268	O
PE3	0.52792	5.61247	7.21333	O
LP3	4.84243	-0.12527	9.56233	O
GEV	34.85383	13.77318	0.14902	O
GUM	34.22850	11.75297	0.00000	O
LGU	0.00000	0.00000	0.00000	X
WBU	7.93970	37.32575	2.32992	O
GPA	0.00000	0.00000	0.00000	X
GLO	40.33837	8.23667	-0.04961	O



1. Preliminary
Analysis

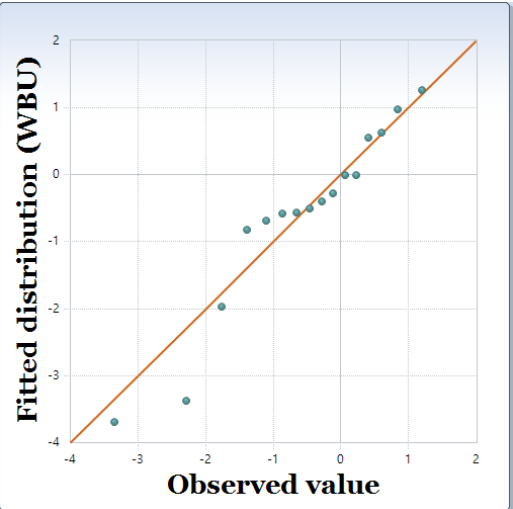
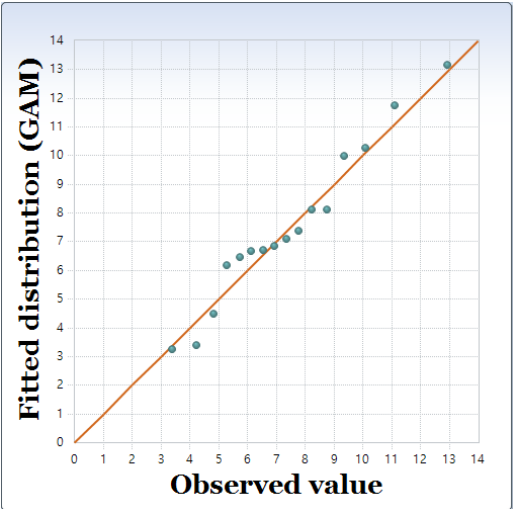
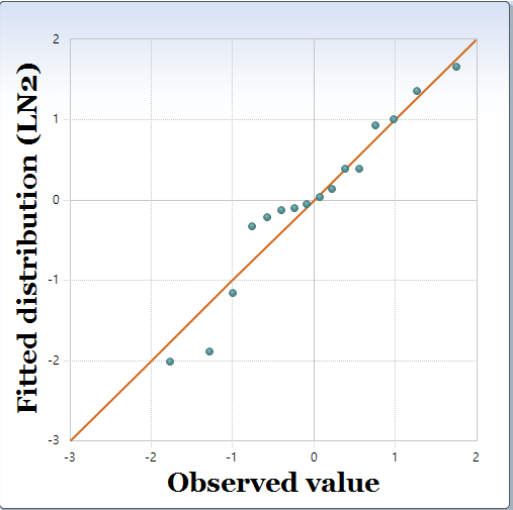
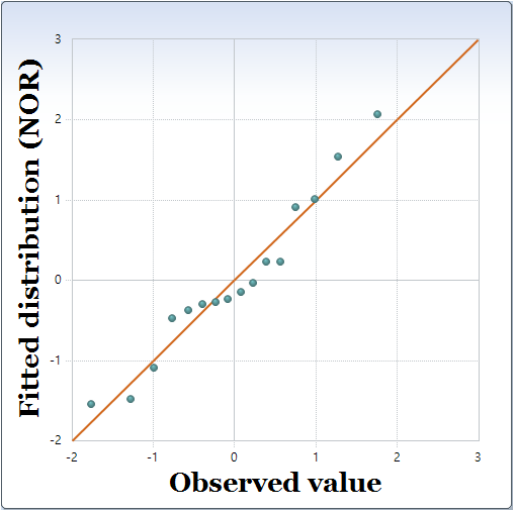
2. Frequency
Analysis

3. Optimization
Modeling

Frequency Analysis

Parameter Estimation : MAXIMUM LIKEIHOOD METHOD

Dist.	LOC PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY
NOR	41.01250	15.07377	0.00000	O
LN2	0.00000	3.64555	0.38168	O
LN3	-38.42094	4.35814	0.18346	O
GAM	0.00000	5.48224	7.48097	O
PE3	-8.25977	4.41446	11.16156	O
LP3	4.78132	-0.13132	8.64909	O
GEV	35.13399	13.31113	0.16335	O
GUM	33.99565	12.77870	0.00000	O
LGU	0.00000	0.00000	0.00000	X
WBU	12.52434	32.09512	2.02726	O
GPA	17.70000	41.79303	0.74498	O
GLO	39.64777	8.16514	-0.11497	O



1. Preliminary Analysis

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Frequency Analysis

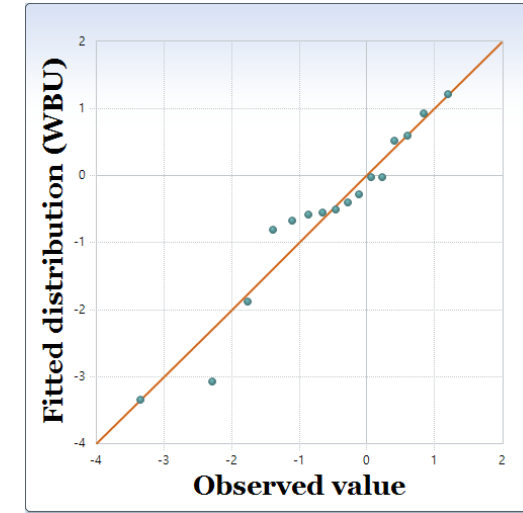
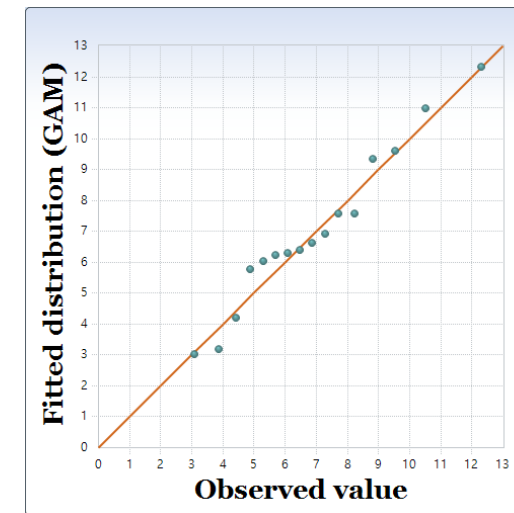
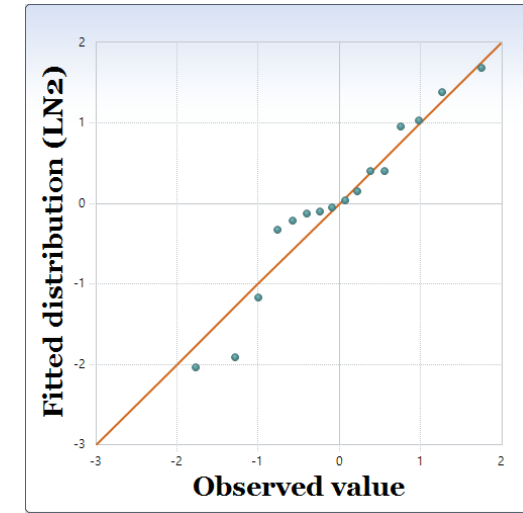
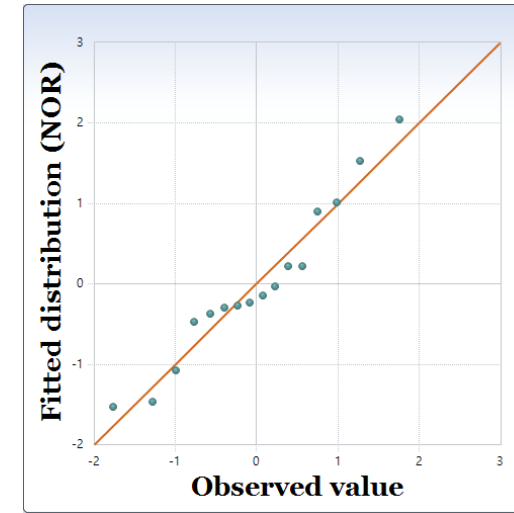
Parameter Estimation : METHOD OF L-MOMENTS

1. Preliminary Analysis

2. Frequency Analysis

3. Optimization Modeling

Dist.	LOC PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY
NOR	41.01250	15.22981	0.00000	O
LN2	0.00000	3.64329	0.37572	O
LN3	-27.51259	4.20230	0.22317	O
GAM	0.00000	5.86103	6.99750	O
PE3	-5.45595	5.13114	9.05617	O
LP3	2.29430	0.11823	11.42948	O
GEV	34.43522	13.46054	0.09744	O
GUM	33.85713	12.39636	0.00000	O
LGU	0.00000	0.00000	0.00000	X
WBU	11.15900	33.69707	2.04211	O
GPA	18.60797	36.01421	0.60745	X
GLO	39.48345	8.42612	-0.10882	O



Frequency Analysis

Goodness-of-fit

METHOD OF MOMENTS

구분	CHI-SQUARED TEST			KOLMOGOROV-SMIRNOV TEST			CRAMER-VON MISES TEST			PROB. PLOT TEST		
	COMPUTED	TABLE	ACCEPT	COMPUTED	TABLE	ACCEPT	COMPUTED	TABLE	ACCEPT	COMPUTED	TABLE	ACCEPT
NOR	7.028	7.815	O	0.161	0.327	O	0.071	0.457	O	0.976	0.940	O
LN2	6.839	7.815	O	0.104	0.327	O	0.069	0.457	O	0.982	0.937	O
LN3	6.684	5.991	X	0.134	0.327	O	0.060	0.457	O	0.983	0.963	O
GAM	6.733	7.815	O	0.117	0.327	O	0.062	0.457	O	0.983	0.954	O
PE3	6.736	5.991	X	0.116	0.327	O	0.063	0.457	O	0.983	0.947	O
LP3	6.903	5.991	X	0.135	0.327	O	0.062	0.457	O	0.982	0.947	O
GEV	6.702	5.991	X	0.132	0.327	O	0.061	0.457	O	0.983	0.963	O
GUM	6.777	7.815	O	0.102	0.327	O	0.067	0.457	O	0.982	0.934	O
LGU	-	2.420	X	-	0.327	X	-	0.457	X	-	0.964	X
WBU	6.999	5.991	X	0.138	0.327	O	0.065	0.457	O	0.982	0.951	O
WAK	-	2.420	X	-	0.327	X	-	0.457	X	-	0.000	X
GPA	-	2.420	X	-	0.327	X	-	0.457	X	-	0.955	X
GLO	6.363	5.991	X	0.131	0.327	O	0.057	0.457	O	0.982	0.957	O

MAXIMUM LIKEIHOOD METHOD

구분	CHI-SQUARED TEST			KOLMOGOROV-SMIRNOV TEST			CRAMER-VON MISES TEST			PROB. PLOT TEST		
	COMPUTED	TABLE	ACCEPT	COMPUTED	TABLE	ACCEPT	COMPUTED	TABLE	ACCEPT	COMPUTED	TABLE	ACCEPT
NOR	7.028	7.815	O	0.161	0.327	O	0.071	0.457	O	0.976	0.940	O
LN2	6.909	7.815	O	0.099	0.327	O	0.075	0.457	O	0.981	0.936	O
LN3	6.666	5.991	X	0.125	0.327	O	0.059	0.457	O	0.983	0.963	O
GAM	6.728	7.815	O	0.116	0.327	O	0.062	0.457	O	0.983	0.954	O
PE3	6.700	5.991	X	0.123	0.327	O	0.060	0.457	O	0.983	0.947	O
LP3	6.939	5.991	X	0.138	0.327	O	0.063	0.457	O	0.981	0.947	O
GEV	6.764	5.991	X	0.130	0.327	O	0.059	0.457	O	0.983	0.964	O
GUM	6.745	7.815	O	0.108	0.327	O	0.068	0.457	O	0.982	0.934	O
LGU	-	2.420	X	-	0.327	X	-	0.457	X	-	0.964	X
WBU	6.968	5.991	X	0.123	0.327	O	0.065	0.457	O	0.982	0.951	O
WAK	-	2.420	X	-	0.327	X	-	0.457	X	-	0.000	X
GPA	8.898	5.991	X	0.170	0.327	O	0.094	0.457	O	0.969	0.962	O
GLO	6.241	5.991	X	0.113	0.327	O	0.052	0.457	O	0.984	0.957	O

METHOD OF L-MOMENTS

구분	CHI-SQUARED TEST			KOLMOGOROV-SMIRNOV TEST			CRAMER-VON MISES TEST			PROB. PLOT TEST		
	COMPUTED	TABLE	ACCEPT	COMPUTED	TABLE	ACCEPT	COMPUTED	TABLE	ACCEPT	COMPUTED	TABLE	ACCEPT
NOR	7.016	7.815	O	0.162	0.327	O	0.071	0.457	O	0.976	0.940	O
LN2	6.905	7.815	O	0.097	0.327	O	0.075	0.457	O	0.981	0.937	O
LN3	6.680	5.991	X	0.122	0.327	O	0.062	0.457	O	0.984	0.962	O
GAM	6.786	7.815	O	0.118	0.327	O	0.065	0.457	O	0.983	0.954	O
PE3	6.748	5.991	X	0.123	0.327	O	0.063	0.457	O	0.983	0.947	O
LP3	7.877	5.991	X	0.111	0.327	O	0.120	0.457	O	0.966	0.947	O
GEV	6.683	5.991	X	0.121	0.327	O	0.062	0.457	O	0.984	0.962	O
GUM	6.779	7.815	O	0.098	0.327	O	0.071	0.457	O	0.982	0.934	O
LGU	-	2.420	X	-	0.327	X	-	0.457	X	-	0.964	X
WBU	7.045	5.991	X	0.128	0.327	O	0.067	0.457	O	0.982	0.951	O
WAK	7.475	2.420	X	0.101	0.327	O	0.055	0.457	O	NaN	0.000	X
GPA	-	2.420	X	-	0.327	X	-	0.457	X	-	0.955	X
GLO	6.229	5.991	X	0.112	0.327	O	0.053	0.457	O	0.984	0.957	O

- NOR, LN2, GAM, GUM 선정

1. Preliminary Analysis

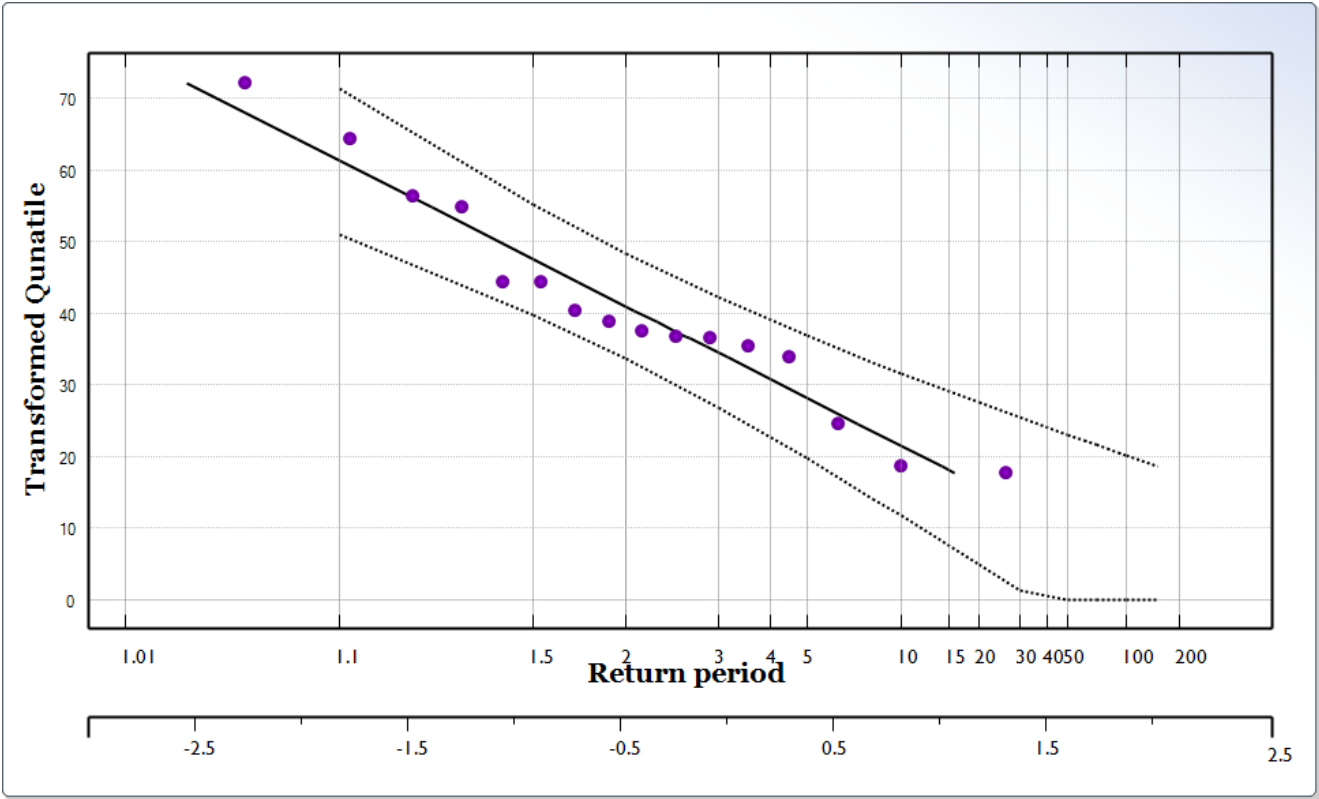
2. Frequency Analysis

3. Optimization Modeling

Frequency Analysis

Result

- Method of L-moments : Normal distribution



Year	Annual Mimimum Storage	Frequency
2014	33.4	4 year
2015	22.1	9 year
2016	25.2	6 year
2017	9.7	50 year
2018	30.1	4 year
2019	31.4	4 year

Return	1.1	1.5	2.0	3.0	5.0	8.0	10	20	30	50	60	70	80	90	100	150
NOR	61.3	47.6	41.0	34.5	28.2	23.5	21.5	16.0	13.1	9.7	8.6	7.7	6.9	6.2	5.6	3.3

1. Preliminary Analysis

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Optimization Modeling

최적화 모델의 종류 선정과 문제설정

- 유입량의 Uncertainty를 반영할 수 있는 SDP(Stochastic Dynamic Programming) 선정

- Objective function

sum of deviation of storage and demand

$$\text{FSD} = \text{Minimize}[(\text{Target storage} - \text{storage})^2 + (\text{Demand} - \text{Release})^2]$$

Target storage = 각 월의 용수공급기준의 평균

Target demand = 각 월의 공급 계약량

- 10 intervals of storage and inflow
- Storage interval = 0~10, 10~20, 20~30, 30~40, 40~50, 50~60, 60~70, 70~80, 80~90, 90~100 (MCM)
- Inflow interval(cumulative) = 0~10%, 10~20%, 20~30%, 30~40%, 40~50%, 50~60%, 60~70%, 70~80%, 80~90%, 90~100%

1. Preliminary
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모델 알고리즘 구축과정

- NDIC-FAT를 사용하여 각 시즌별로 확률을 주기위해 정상성을 확인한 1998~2013년도의 유입량을 분포에 피팅시켜 파라미터 계산

1월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	1.49937	1.23414	0.00000	O	
LN2	0.00000	0.02132	0.87605	O	
LN3	-1.40659	0.97376	0.43129	O	
GAM	0.00000	1.24356	1.20572	O	
PE3	-0.55660	0.81776	2.51414	O	
LP3	-1.70674	1.31754	1.19668	X	
GEV	0.89323	0.94798	-0.05942	O	
GUM	0.91954	1.00454	0.00000	O	
LGU	-15.06180	0.89323	16.83058	O	
WBU	-0.31171	1.99315	1.42841	O	
GPA	-0.10863	2.10551	0.30939	O	
GLO	1.26544	0.64747	-0.20868	O	

2월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	2.60562	2.41519	0.00000	O	
LN2	0.00000	0.45206	1.00560	O	
LN3	-1.19745	1.11929	0.65806	O	
GAM	0.00000	2.93181	0.88874	O	
PE3	-0.24981	2.53489	1.12645	O	
LP3	0.00000	0.00000	0.00000	X	
GEV	1.30461	1.54806	-0.21255	O	
GUM	1.47090	1.96585	0.00000	O	
LGU	-5.97852	1.30461	4.70469	O	
WBU	-0.21852	2.88174	1.03225	O	
GPA	-0.17966	2.90801	0.04406	O	
GLO	1.93521	1.15209	-0.31403	O	

3월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	2.96250	2.43387	0.00000	O	
LN2	0.00000	0.70394	0.87418	O	
LN3	-1.63889	1.37952	0.54192	O	
GAM	0.00000	2.44563	1.21134	O	
PE3	-0.38627	2.05820	1.62704	O	
LP3	-1.51650	0.57400	3.80842	O	
GEV	1.70553	1.71746	-0.13639	O	
GUM	1.81900	1.98106	0.00000	O	
LGU	-10.88678	1.70553	7.33196	O	
WBU	-0.20648	3.38323	1.22043	O	
GPA	-0.02140	3.50014	0.17301	O	
GLO	2.39321	1.22479	-0.26063	O	

- GEV distribution 선정

4월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	7.04625	7.20399	0.00000	O	
LN2	0.00000	1.31103	1.13267	O	
LN3	-0.99851	1.81838	0.96581	O	
GAM	0.00000	10.41182	0.67675	O	
PE3	0.50768	12.11345	0.53978	X	
LP3	-2.30449	0.55182	6.49475	O	
GEV	2.87688	3.46699	-0.39229	O	
GUM	3.66162	5.86371	0.00000	O	
LGU	-5.96096	2.87688	2.54914	O	
WBU	0.28706	5.70024	0.75376	X	
GPA	-0.11455	5.45530	-0.23817	O	
GLO	4.33257	2.84783	-0.44832	O	

5월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	8.32125	7.37104	0.00000	O	
LN2	0.00000	1.66438	0.95335	O	
LN3	-12.51813	2.97295	0.35748	O	
GAM	0.00000	8.31112	1.00122	O	
PE3	-6.17244	4.01675	3.60831	O	
LP3	-0.49167	2.05373	0.88834	X	
GEV	4.84384	5.96851	-0.00561	O	
GUM	4.85814	5.99969	0.00000	O	
LGU	-1.05885060	4.84284	178.21758	O	
WBU	-3.53167	13.22335	1.60413	O	
GPA	-1.69491	14.10778	0.40850	O	
GLO	7.15165	3.95570	-0.17354	O	

6월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	11.95437	10.99859	0.00000	O	
LN2	0.00000	1.98432	0.99677	O	
LN3	-4.09863	2.69523	0.62942	O	
GAM	0.00000	13.18734	0.90651	O	
PE3	-1.49578	10.97585	1.22543	O	
LP3	-2.82623	0.32106	14.85149	O	
GEV	6.08559	7.22356	-0.19421	O	
GUM	6.78694	8.95234	0.00000	O	
LGU	-31.10988	6.08559	5.14919	O	
WBU	-1.23230	13.62429	1.08992	O	
GPA	-0.91909	13.83376	0.07459	O	
GLO	9.01659	5.32107	-0.30098	O	

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모델 알고리즘 구축과정

- NDIC-FAT를 사용하여 각 시즌별로 확률을 주기위해 정상성을 확인한 1998~2013년도의 유입량을 분포에 피팅시켜 파라미터 계산

7월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC. PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	43.62375	25.61137	0.00000	O	
LN2	0.00000	3.59255	0.60506	O	
LN3	0.86295	3.56462	0.61807	O	
GAM	0.00000	16.52052	2.64058	O	
PE3	11.96278	25.04126	1.26835	O	
LP3	-1.00978	0.07845	58.98184	O	
GEV	30.01110	16.98193	-0.18686	O	
GUM	31.59085	20.84646	0.00000	O	
LGU	-60.86989	30.01110	5.35163	O	
WBU	12.60346	32.20062	1.10548	O	
GPA	13.46855	32.77609	0.08691	O	
GLO	36.89051	12.45802	-0.29579	O	

8월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC. PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	37.31437	28.08623	0.00000	O	
LN2	0.00000	3.30555	0.79225	O	
LN3	-26.13030	4.04884	0.45018	O	
GAM	0.00000	24.91077	1.49792	O	
PE3	-7.78170	19.47061	2.31611	O	
LP3	-0.28519	0.23025	15.60799	O	
GEV	23.39036	21.27378	-0.07287	O	
GUM	24.11871	22.86089	0.00000	O	
LGU	-268.55626	23.39036	13.72331	O	
WBU	-3.01276	44.18944	1.38874	O	
GPA	1.10476	46.53301	0.28510	O	
GLO	31.77312	14.64008	-0.21762	O	

9월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC. PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	22.37375	21.05527	0.00000	O	
LN2	0.00000	2.56989	1.02391	O	
LN3	-29.06160	3.85412	0.41524	O	
GAM	0.00000	26.22038	0.85330	O	
PE3	-13.87645	13.40751	2.70372	O	
LP3	0.09734	2.48450	0.82534	X	
GEV	12.11723	16.36398	-0.04789	O	
GUM	12.48142	17.13801	0.00000	O	
LGU	-329.59482	12.11723	20.88196	O	
WBU	-9.11734	34.77508	1.46367	O	
GPA	-5.30895	36.82789	0.33036	O	
GLO	18.52236	11.10476	-0.20107	O	

- GEV distribution 선정

10월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC. PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	5.52688	6.68134	0.00000	O	
LN2	0.00000	0.71231	1.41231	O	
LN3	-0.19659	0.83996	1.34552	O	
GAM	0.00000	14.00371	0.39467	O	
PE3	0.49376	18.52816	0.27165	X	
LP3	-3.71577	0.66923	6.58826	O	
GEV	1.32159	2.13645	-0.57267	O	
GUM	2.38780	5.43830	0.00000	O	
LGU	-2.20910	1.52159	1.74620	O	
WBU	0.31404	2.97287	0.53975	X	
GPA	-0.15390	2.88026	-0.49298	O	
GLO	2.42919	1.92480	-0.59552	O	

11월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC. PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	1.57313	1.38303	0.00000	O	
LN2	0.00000	0.00653	0.94502	O	
LN3	-0.55310	0.52636	0.87527	O	
GAM	0.00000	1.54095	1.02088	O	
PE3	-0.03073	1.49513	1.07272	O	
LP3	-1.95925	0.91555	2.04622	X	
GEV	0.82405	0.87342	-0.22344	O	
GUM	0.92334	1.12572	0.00000	O	
LGU	-3.08488	0.82405	4.47542	O	
WBU	-0.02066	1.61348	1.03065	O	
GPA	-0.00782	1.62220	0.02610	O	
GLO	1.18065	0.65398	-0.32183	O	

12월

METHOD OF MOMENTS	MAXIMUM LIKELIHOOD METHOD	METHOD OF L-MOMENTS			
DISTRIBUTION	LOC. PAR.	SCALE PAR.	SHAPE PAR.	VALIDITY	
NOR	1.89375	1.83286	0.00000	O	
LN2	0.00000	0.07779	1.59403	O	
LN3	-0.23383	0.32899	0.92303	O	
GAM	0.00000	2.39609	0.79025	O	
PE3	0.18193	2.90207	0.58996	X	
LP3	-2.75002	0.65052	4.27440	X	
GEV	0.84050	0.92030	-0.36917	O	
GUM	1.03262	1.49187	0.00000	O	
LGU	-1.65242	0.84050	2.70881	O	
WBU	0.12864	1.53841	0.78631	X	
GPA	0.03604	1.47964	-0.20352	X	
GLO	1.22577	0.74660	-0.43037	O	

1. Preliminary
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Optimization Modeling

모델 알고리즘 구축과정

- Transition probability matrix
- 전이 확률 행렬의 경우 어떤식으로 산정하는지 확실치 않아 난수 발생을 통해 행렬 생성

```
#Transition probability (one-step)
jan.to.feb = matrix(nrow=10, ncol=10, byrow=T)
feb.to.mar = matrix(nrow=10, ncol=10, byrow=T)
mar.to.apr = matrix(nrow=10, ncol=10, byrow=T)
apr.to.may = matrix(nrow=10, ncol=10, byrow=T)
may.to.jun = matrix(nrow=10, ncol=10, byrow=T)
jun.to.jul = matrix(nrow=10, ncol=10, byrow=T)
jul.to.aug = matrix(nrow=10, ncol=10, byrow=T)
aug.to.sep = matrix(nrow=10, ncol=10, byrow=T)
sep.to.oct = matrix(nrow=10, ncol=10, byrow=T)
oct.to.nov = matrix(nrow=10, ncol=10, byrow=T)
nov.to.dec = matrix(nrow=10, ncol=10, byrow=T)
dec.to.jan = matrix(nrow=10, ncol=10, byrow=T)

Month.transition.prob = list(jan.to.feb, feb.to.mar, mar.to.apr, apr.to.may, may.to.jun, jun.to.jul,
                             jul.to.aug, aug.to.sep, sep.to.oct, oct.to.nov, nov.to.dec, dec.to.jan)

for (month in 1:12){
  for (i in 1:10){
    x = runif(10,0,1)
    y = x / sum(x)
    Month.transition.prob[[month]][i,1] = y[1]
    Month.transition.prob[[month]][i,2] = y[2]
    Month.transition.prob[[month]][i,3] = y[3]
    Month.transition.prob[[month]][i,4] = y[4]
    Month.transition.prob[[month]][i,5] = y[5]
    Month.transition.prob[[month]][i,6] = y[6]
    Month.transition.prob[[month]][i,7] = y[7]
    Month.transition.prob[[month]][i,8] = y[8]
    Month.transition.prob[[month]][i,9] = y[9]
    Month.transition.prob[[month]][i,10] = y[10]
  }
}
```

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모델 알고리즘 구축과정

- FSD dataframe

```
> Month.fsd[[12]][800:825,]
```

	k	st	i	Qt	l	st+1	Rt	FSD
[1,]	8	75	10	0.1873532762	10	95	NA	NA
[2,]	9	85	1	0.0003524752	1	5	8.000035e+01	6312.75088
[3,]	9	85	1	0.0003524752	2	15	7.000035e+01	4833.38928
[4,]	9	85	1	0.0003524752	3	25	6.000035e+01	3552.02768
[5,]	9	85	1	0.0003524752	4	35	5.000035e+01	2468.66607
[6,]	9	85	1	0.0003524752	5	45	4.000035e+01	1583.30447
[7,]	9	85	1	0.0003524752	6	55	3.000035e+01	895.94287
[8,]	9	85	1	0.0003524752	7	65	2.000035e+01	406.58127
[9,]	9	85	1	0.0003524752	8	75	1.000035e+01	115.21966
[10,]	9	85	1	0.0003524752	9	85	3.524752e-04	21.85806
[11,]	9	85	1	0.0003524752	10	95	NA	NA
[12,]	9	85	2	0.0102037464	1	5	8.001020e+01	6314.30587
[13,]	9	85	2	0.0102037464	2	15	7.001020e+01	4834.74921
[14,]	9	85	2	0.0102037464	3	25	6.001020e+01	3553.19255
[15,]	9	85	2	0.0102037464	4	35	5.001020e+01	2469.63589
[16,]	9	85	2	0.0102037464	5	45	4.001020e+01	1584.07923
[17,]	9	85	2	0.0102037464	6	55	3.001020e+01	896.52258
[18,]	9	85	2	0.0102037464	7	65	2.001020e+01	406.96592
[19,]	9	85	2	0.0102037464	8	75	1.001020e+01	115.40926
[20,]	9	85	2	0.0102037464	9	85	1.020375e-02	21.85260
[21,]	9	85	2	0.0102037464	10	95	NA	NA
[22,]	9	85	3	0.0179967699	1	5	8.001800e+01	6315.53610
[23,]	9	85	3	0.0179967699	2	15	7.001800e+01	4835.82514
[24,]	9	85	3	0.0179967699	3	25	6.001800e+01	3554.11418
[25,]	9	85	3	0.0179967699	4	35	5.001800e+01	2470.40322
[26,]	9	85	3	0.0179967699	5	45	4.001800e+01	1584.69226

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Optimization Modeling

모델 알고리즘 구축과정

- Stochastic dynamic programming with backward recursive equation

```
#First step (T=12)
m.fsd = back.Month.fsd[[1]]
ft.r = data.frame()
for (i in 1:10){
  #i=1
  outer = m.fsd[which(m.fsd[,1]==i),]
  back.k = list()
  back.i = list()
  back.ft = list()
  back.l = list()
  back.r = list()
  for (j in 1:10){
    #j=1
    inner = outer[which(outer[,3]==j),]
    ft = min(inner[,8],na.rm=T)
    ft.loc = which(inner[,8]==ft)
    rt = inner[ft.loc,7]

    back.k[[j]] = i
    back.i[[j]] = j
    back.ft[[j]] = ft
    back.l[[j]] = ft.loc
    back.r[[j]] = rt
  }
  mid = data.frame(cbind(back.k), cbind(back.i), cbind(back.ft), cbind(back.l), cbind(back.r))
  ft.r = rbind(ft.r, mid)
}
ft.result[[1]][[1]] = ft.r
```

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Optimization Modeling

모델 알고리즘 구축과정

- Stochastic dynamic programming with backward recursive equation (Continue)

```
#Backward recursive process
for (i in 2:2000){
  #i=2
  step = i%%12
  if (step==0){
    step=12
  }
  stepback = step - 1
  if (stepback==0){
    stepback=12
  }

  m.fsd = back.Month.fsd[[step]]
  m.trans.prob = back.Month.transition.prob[[step]]
  ll = length(ft.result[[step]])
  lll = length(ft.result[[stepback]])

  back.step.result = ft.result[[stepback]][lll]

  ft.r = data.frame()
  for (j in 1:10){
    #j=1
    outer = m.fsd[which(m.fsd[,1]==j),]
    back.k = list()
    back.i = list()
    back.ft = list()
    back.l = list()
    back.r = list()
    for (k in 1:10){
      #k=1
      inner = outer[which(outer[,3]==k),]
      for (l in 1:10){
        #l=1
        if (is.na(inner[l,8])){
          next
        }
        dd = back.step.result[which(back.step.result[,1]==1),]

        inner[l,8] = inner[l,8] + ((m.trans.prob[k,1] * dd[1,3][[1]]) + (m.trans.prob[k,2] * dd[2,3][[1]]) +
          (m.trans.prob[k,3] * dd[3,3][[1]]) + (m.trans.prob[k,4] * dd[4,3][[1]]) +
          (m.trans.prob[k,5] * dd[5,3][[1]]) + (m.trans.prob[k,6] * dd[6,3][[1]]) +
          (m.trans.prob[k,7] * dd[7,3][[1]]) + (m.trans.prob[k,8] * dd[8,3][[1]]) +
          (m.trans.prob[k,9] * dd[9,3][[1]]) + (m.trans.prob[k,10] * dd[10,3][[1]]))

      }
    }

    ft = min(inner[,8],na.rm=T)
    ft.loc = which(inner[,8]==ft)
    rt = inner[ft.loc,7]
  }
}
```

1. Preliminary
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모델 알고리즘 구축과정

- Stochastic dynamic programming with backward recursive equation (Continue)

```
> ft.result[[12]][[9]][1:25,]
      back.k back.i back.ft back.l      back.r
1           1      1 1043.054      1          0
2           1      2 1043.043      1 0.009589822
3           1      3 1043.036      1  0.0189134
4           1      4 1043.039      1  0.02732173
5           1      5 1043.04      1  0.0357333
6           1      6 1043.026      1  0.04477945
7           1      7 1043.02      1  0.05521051
8           1      8 1043.01      1  0.06834215
9           1      9 1043.007      1  0.08745786
10          1     10 1042.996      1  0.1280816
11          2      1  520.7538      2          0
12          2      2  520.7435      2 0.009589822
13          2      3  520.7363      2  0.0189134
14          2      4  520.739      2  0.02732173
15          2      5  520.7396      2  0.0357333
16          2      6  520.7261      2  0.04477945
17          2      7  520.7195      2  0.05521051
18          2      8  520.7101      2  0.06834215
19          2      9  520.7074      2  0.08745786
20          2     10  520.6958      2  0.1280816
21          3      1  214.4538      3          0
22          3      2  214.4435      3 0.009589822
23          3      3  214.4363      3  0.0189134
24          3      4  214.439      3  0.02732173
25          3      5  214.4396      3  0.0357333
```

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모델 알고리즘 구축과정

- Running with observed inflows and initial storage

```
inflow = read.csv(file=paste0(path,'inflow.csv'), header = T)
colnames(inflow) = c('Year', 'Month', 'Date', 'Inflow')
storage.result = array(NA, c(1765))
storage.result[1] = 45.7
min.rule = read.csv(file=paste0(path,'minimumrule.csv'), header = F)
min.rule[1,1] = 'Jan'
for (i in 1:1764){
  #i=1
  inflows = inflow[i,4]
  storages = storage.result[i]
  month = inflow[i,2]
  op.rule = optimal.rule[[month]]
  boundaries = Month.inflow[[month]]
  min.rules = min.rule[month,2]

  if (storages < 10 & storages > 0){
    st.loc = 1
  }else if (storages < 20 & storages > 10){
    st.loc = 2
  }else if (storages < 30 & storages > 20){
    st.loc = 3
  }else if (storages < 40 & storages > 30){
    st.loc = 4
  }else if (storages < 50 & storages > 40){
    st.loc = 5
  }else if (storages < 60 & storages > 50){
    st.loc = 6
  }else if (storages < 70 & storages > 60){
    st.loc = 7
  }else if (storages < 80 & storages > 70){
    st.loc = 8
  }else if (storages < 90 & storages > 80){
    st.loc = 9
  }else{
    st.loc = 10
  }

  if (inflows < boundaries[2] & inflows > boundaries[1]){
    in.loc = 1
  }else if (inflows < boundaries[3] & inflows > boundaries[2]){
```

```
    if (inflows < boundaries[2] & inflows > boundaries[1]){
      in.loc = 1
    }else if (inflows < boundaries[3] & inflows > boundaries[2]){
      in.loc = 2
    }else if (inflows < boundaries[4] & inflows > boundaries[3]){
      in.loc = 3
    }else if (inflows < boundaries[5] & inflows > boundaries[4]){
      in.loc = 4
    }else if (inflows < boundaries[6] & inflows > boundaries[5]){
      in.loc = 5
    }else if (inflows < boundaries[7] & inflows > boundaries[6]){
      in.loc = 6
    }else if (inflows < boundaries[8] & inflows > boundaries[7]){
      in.loc = 7
    }else if (inflows < boundaries[9] & inflows > boundaries[8]){
      in.loc = 8
    }else if (inflows < boundaries[10] & inflows > boundaries[9]){
      in.loc = 9
    }else{
      in.loc = 10
    }

    opp.rule = op.rule[which(op.rule[,1]==st.loc),]
    oppp.rule = opp.rule[which(opp.rule[,2]==in.loc),]
    op.release = oppp.rule[,5][[1]]
    storage.result[i+1] = max(min.rules, storages + inflows - op.release)
  }
}
```

1. Preliminary
Analysis

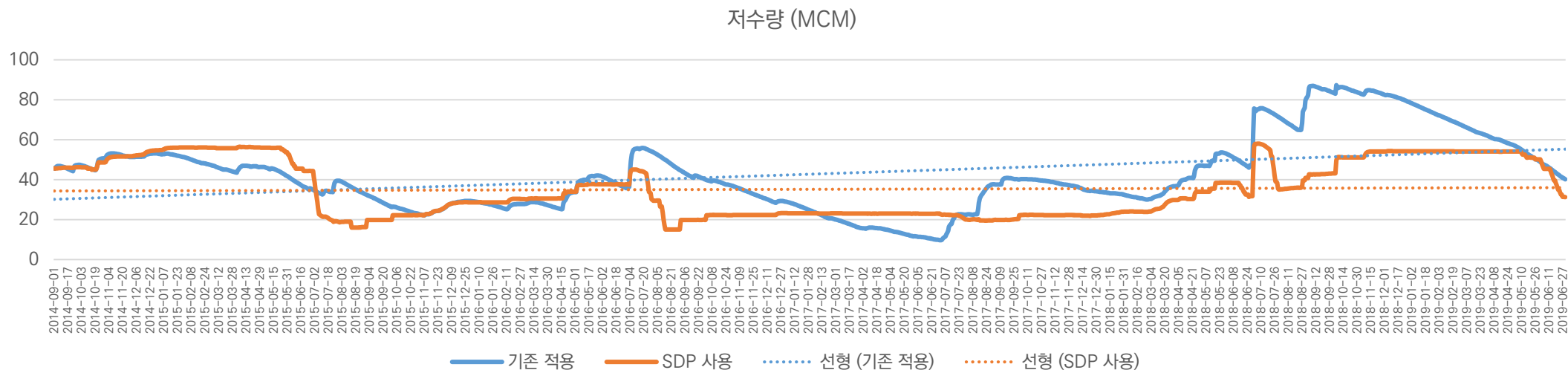
2. Frequency
Analysis

3. Optimization
Modeling

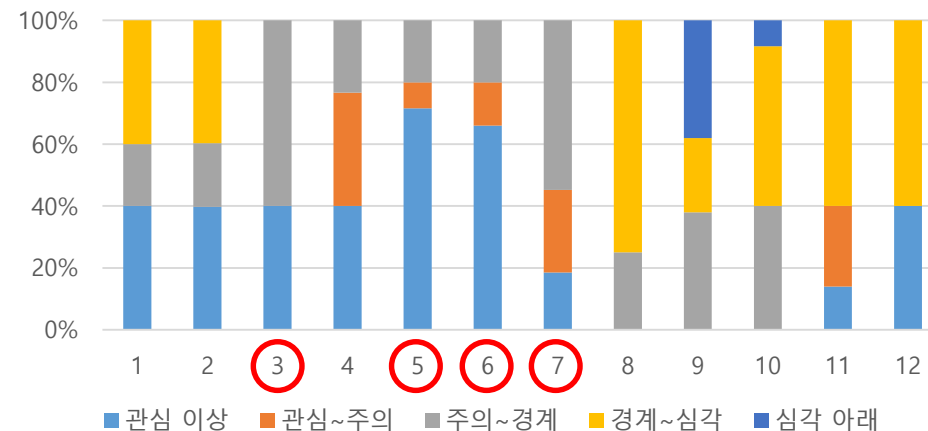
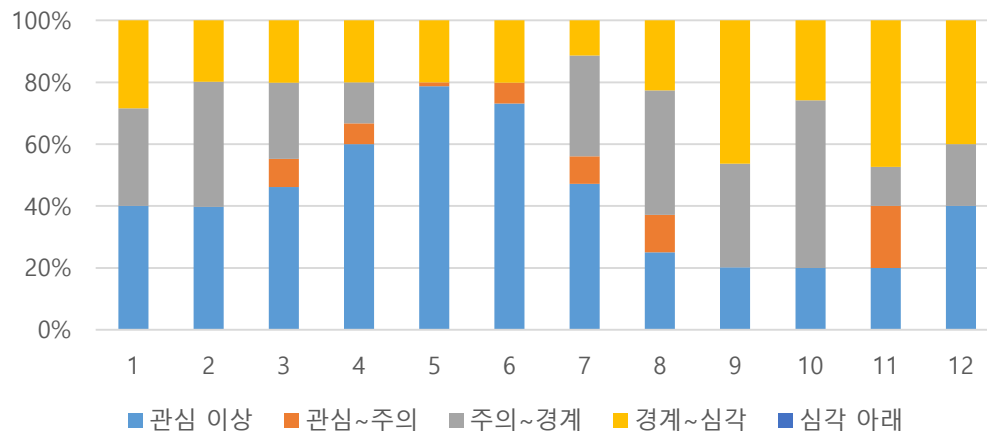
Optimization Modeling

Result

- 2014년 9월 1일부터 2019년 6월 30일까지의 일별 저수지 모의



- 실제 운영 실적과 비교하여 최적 운영률이 어떤 측면에서 얼마나 개선된 결과를 보였는가?



Optimization Modeling

Discussion – Future study

- Development of transition probability matrix
 - 실제 관측치를 구간별로 나누어서 correlation test 진행
- 목적함수 재설정
 - 유입량과 저수량을 사용한 실제 상황을 더 반영할 수 있는 목적함수 설정
 - 저수량 구간별 크기와 유입량 구간별 크기가 많이 차이 남
- Interval의 세분화
 - 유입량에 비해 저수량의 interval 단위가 너무 커서 demand에 비해 너무 큰 방류량이 계산됨
 - 저수량을 더욱 잘게 분할하여 더 세밀한 stochastic dynamic programming이 가능

1. Preliminary
Analysis

2. Frequency
Analysis

3. Optimization
Modeling



Thank you